TSA 2009

Problem Solving

Questions taken from the TSA Oxford 2009 Section 1 question paper

Questions	Total marks	Time limit
25	25	45 minutes

Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Questions are numbered 1 to 25 for this set. The number in brackets shows the question's number in the original TSA 2009 paper, e.g. **1 (Q3)**.

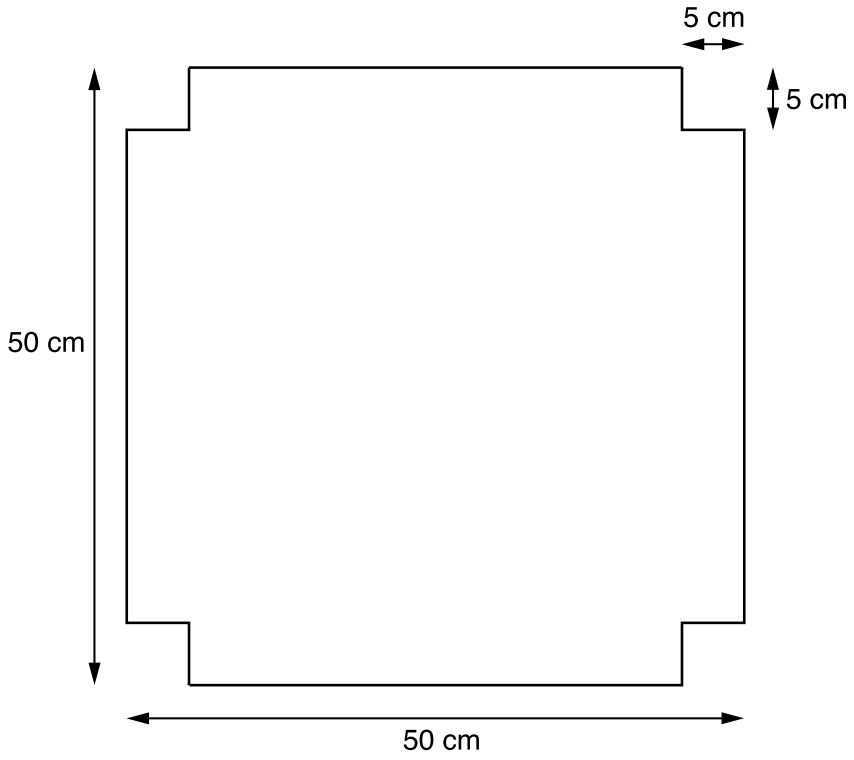
How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to the 25 questions in this set: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes).

Score: _____ / 25

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1 (Q2) A manufacturer wishes to make an open-topped box out of the piece of cardboard shown below by folding up its sides.



What is the volume of this box in cubic centimetres?

A 1 600
B 2 400
C 8 000
D 10 125
E 12 500

2 (Q6) The petrol tank of Jenny's car holds 50 litres of petrol. A warning light comes on when only 5 litres are left. She always fills it as soon as she can after the warning light comes on. The car travels 100km on 8 litres of petrol. Last time she put petrol in, which she was able to do immediately after the light came on, she only had $6 with her so she spent it all on fuel at 60c per litre. She then drove 50km and saw a new garage offering petrol at 50c per litre - an offer too good to miss. She filled the tank completely and paid with her credit card.

How much did it cost?

A $ 15.50
B $ 17.50
C $ 19.50
D $ 22.50
E $ 24.50

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3 (Q7) The following table gives figures for the percentage growth per year of labour productivity per person per year in various countries during three periods.

	Period 1	*Period 2*	*Period 3*
Japan	8.5	3.0	3.2
France	5.4	3.0	2.6
United Kingdom	3.6	1.5	2.4
Belgium	3.3	2.8	2.3
Sweden	4.1	1.5	1.8
Denmark	4.3	2.6	1.7
Italy	6.3	3.0	1.6
Netherlands	4.8	2.7	1.6
Germany	4.5	3.1	1.6
United States	2.2	0.0	0.8

Which country's percentage growth per year remained consistently greater than half of its Period 1 level throughout the following periods?

A France

B United Kingdom

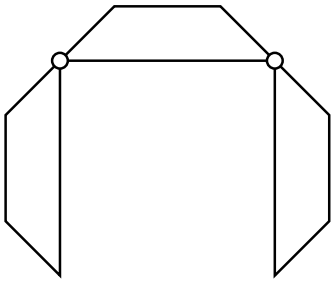
C Germany

D Denmark

E Belgium

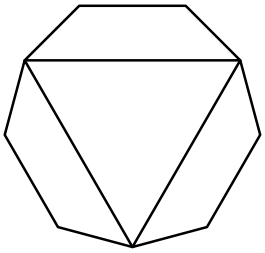
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4 (Q8) The diagram below shows a rather unusual conference table which may take up a range of configurations. The three pieces are hinged together at the points shown by the circles:

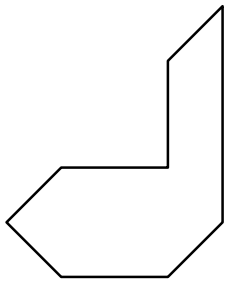


Which of the following shapes could **NOT** be made from this table?

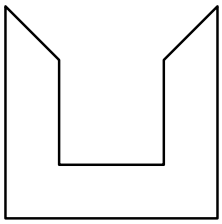
A



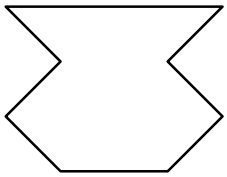
B



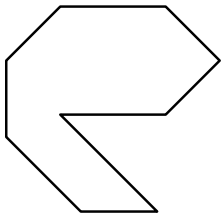
C



D



E



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5 (Q12) Ionnais lives at the bottom of a very steep hill. Worse than that, the post office is at the top. Every Thursday he must go there to collect his pension. He can only walk at 2 km per hour uphill but manages 4 km per hour downhill. The round trip, excluding the wait at the post office, takes him 4½ hours.

How far is it from his house to the post office?

A 3.00 km
B 6.00 km
C 6.75 km
D 12.00 km
E 13.50 km

6 (Q13) Telephone network operators offer customers a choice of tariffs, with the tariff broken into two parts: (a) a fixed monthly rental charge regardless of use, and (b) a charge per minute of call time used. The table below shows the tariffs offered by five companies; all figures are given in £. I am considering taking a contract with one of these and estimate that I will use 100 minutes of call-time each month.

	High user tariff		*Medium user tariff*		*Low user tariff*	
	Rental	Call charge	Rental	Call charge	Rental	Call charge
Activering	50	0.05	20	0.30	0	0.60
Bellaphone	55	0.05	20	0.25	0	0.60
Canconnect	60	0.05	20	0.35	0	0.55
Dialfast	65	0.05	10	0.45	5	0.45
Engaged	70	0.05	10	0.40	0	0.70

Which company should I use to receive the lowest telephone bill?

A Activering
B Bellaphone
C Canconnect
D Dialfast
E Engaged

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7 (Q14) 100 pellets of chicken food will feed one large chicken and two small chickens for a day. 175 pellets of food will feed two large and three small chickens for a day.

Which of the following statements is **not** true?

A A small chicken can be fed for longer than a day with 30 pellets of food.

B 70 pellets of food are sufficient to feed a large chicken for a day.

C A large chicken needs twice as many pellets as a small chicken.

D Two large and four small chickens will need 200 pellets of food for a day.

E Three large and ten small chickens will need 450 pellets of food for a day.

8 (Q18) A boy is given $1.00 by his grandparents to buy sweets. He decides to spend at least half his money on gobstoppers at 5c each, at least a quarter of his money on fruit chews at 3c each and at least one tenth of his money on pieces of bubble gum at 2c each. He will decide how to spend the rest of the money when he gets to the shop.

What is the possible range of number of pieces of bubble gum he can buy?

A From 5 to 9

B From 5 to 11

C From 5 to 13

D From 6 to 12

E From 10 to 22

9 (Q19) My friend's grandmother went on an overseas trip last year visiting several countries. When he visited her this summer she claimed she had not been able to buy him a gift as every country she visited was having a public holiday while she was there. He was very suspicious and consulted his diary to find out whether this was possible.

International Public Holidays:

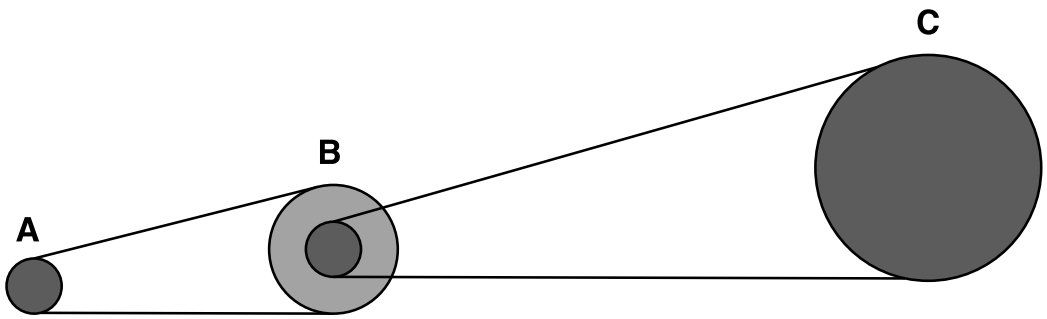
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Denmark	1			9,10,13	8,21	1,5						25,26
Finland	1,6			10,13	1,21,31	20				31		6,25,26
France	1			13	1,8,21	1	14	15			1,11	25
Germany	1,6			10,13	1,21	1,11		15		3,31	1	25,26
Greece	1,6		2,25	17,20	1	8		15		28		25,26
Hong Kong	1,28,29,30			5,10,13	30	13,15		29,31	5			25,26, 31
Israel		11	12	11,17,30	24,31			2	21,22,30	5		14
Italy	1,6			12,13,25	1			15			1	8,25,26

What was the longest time she could have been away?

A 2 days
B 3 days
C 4 days
D 5 days
E 6 days

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10 (Q20) The diagram shows three pulleys driven by belts. Pulley A and the small, inner pulley at B are 1 m across, the outer pulley at B is 2 m across and pulley C is 4 m across.



Pulley A rotates at a rate of 8 complete turns every second. If the belts do not slip, then how many complete turns every second will pulley C make?

A 1
B 4
C 8
D 16
E 64

11 (Q24) Hockey is an 11-a-side game, but a team may consist of up to 16 players, with unlimited substitutions allowed throughout a match of 70 minutes duration.

Roger captains the Buccaneers hockey team. He has 16 players for today's match. He will play the whole match himself, as will the goalkeeper. He intends to rotate all the others in such a way that each of them spends the same total amount of time on the pitch.

How much time should each player (except Roger and the goalkeeper) spend on the pitch during today's match?

A 39 3/8 minutes
B 45 minutes
C 46 2/3 minutes
D 48 1/8 minutes
E 55 minutes

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12 (Q25) Julie lives in London. She has three daughters; April, who lives in New York (where the local time is 5 hours behind London), May, who lives in Vancouver (8 hours behind London), and June, who lives in Tokyo (9 hours ahead of London). They all communicate with each other regularly.

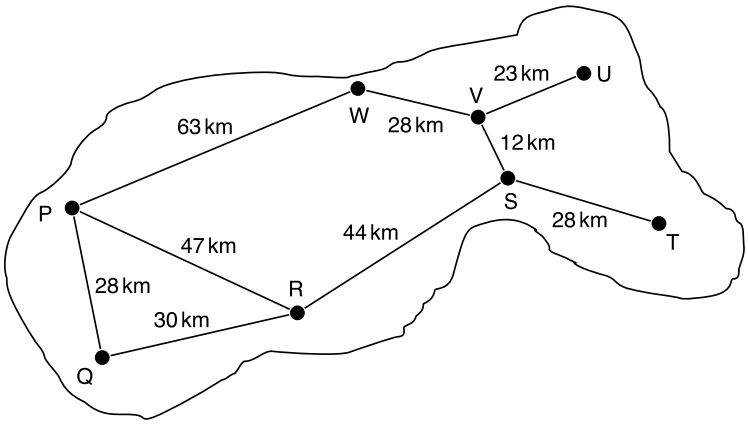
Last week April received a text message from May that said "call me on Wednesday at 23.30 your time." Unfortunately, the message had been sent to April by mistake and was intended for June.

How much later than expected did May's phone eventually ring?

A 12 hours
B 13 hours
C 14 hours
D 17 hours
E 22 hours

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13 (Q26) This is a map of the island of Metallica showing the eight main towns (labelled P - W) and the roads that connect them, together with a chart giving the shortest distance (in km) by road between each pair of towns:



Copper	Gold	Iron	Lead	Nickel	Silver	Tin	Zinc
74							
28	91						
86	12	91					
102	28	119	40				
91	40	63	28	68			
109	35	114	23	63	51		
30	44	47	56	72	84	79	

Which of the towns on the map is Nickel?

A Q

B R

C S

D T

E W

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14 (Q30) In a particular year the month of January (which has 31 days) contains five Fridays.

Which one of the following could **NOT** be true?

A The first Friday in the month occurs before the 4th.

B The 1st of January is a Wednesday.

C There are only 4 Tuesdays.

D There are also 5 Sundays.

E The 31st of January is a Monday.

15 (Q31) A small restaurant is open for 45 weeks each year. It employs 6 people at total wages of £1200 per week (they are not paid when the restaurant is closed). It is open for lunch on Monday to Friday and dinner on Tuesday to Saturday and on average 20 people eat at each opening. The owners pay rental on the premises of £36 000 per year. Other expenses (insurance, gas electricity, water, rates etc.) come to £9000 per year. The cost of buying food works out at £5 per meal served.

How much must the owner charge for each meal to cover expenses?

A £ 10
B £ 11
C £ 16
D £ 22
E £ 27

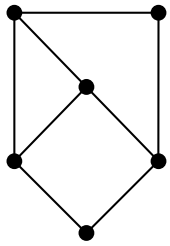
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16 (Q32) Boland City has just installed a new tram system. Several artists have been asked to come up with a map of the system which, whilst it may not show the exact relative positions of the stations, shows the connections between them correctly.

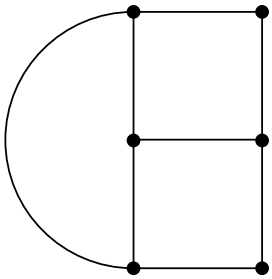
The five maps are shown below (the names of the stations are not marked).

Four of the artists drew possible maps, but the fifth got the connections wrong. Which map below is based on incorrect connections?

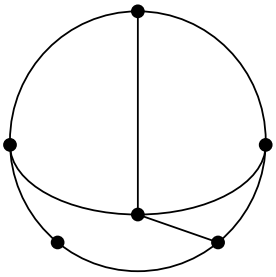
A



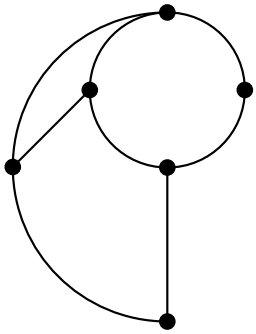
B



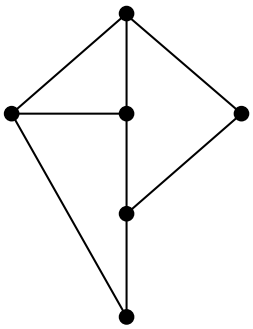
C



D



E



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17 (Q36) The rowing machine in our gym has a display that tells the user how much work they have done. The display shows the following information:

1	distance rowed since starting	
2	number of calories used per 500 m rowing	(an average update every 15 seconds)
3	time since starting	
4	number of rowing strokes per minute	(an average update every 30 seconds)

Today I have used the machine three times, each time as part of a circuit of training using several machines in the gym. During my third session, I re-set the counter after rowing 500 m.

	distance rowed	*number of calories used per 500 m*
first	400 m	50
second	600 m	55
third	800 m	60 for the first 500 m
		40 for remainder

How many calories did I use up on the rowing machine?

A 155
B 190
C 205
D 206
E 260

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18 (Q37) When Mrs Brown looks out of her back room window, a bungalow 7m high and 30m beyond her garden fence is just obscured by a shrub in her hedge which is 1.6m high. She wants to choose a tree to plant beside the shrub which, when fully grown, will have the same effect on a small block of flats 20m high beside the bungalow. However, she does not want it to be much bigger than necessary or it will block out her light.

The characteristics of some trees are shown below:

	Characteristics			
Tree	*Features*	*Growth*	*Height (m)*	*Width (m)*
Copper Beech	Copper red leaves	average	20.0	13.0
Flowering cherry	Beautiful blossom	vigorous	10.0	8.0
Hawthorn	Rugged and sturdy	average	8.0	5.2
Holly	Shiny green leaves	slow	7.0	3.2
Hornbeam	Silvery leaves	average	4.8	3.8
Japanese Maple	Red autumn leaves	slow	4.0	4.2
Mountain Ash	Bright red berries	average	4.5	2.5
Sycamore	Dense foliage	vigorous	13.5	9.5

Which of the trees above would be her best choice?

A Copper Beech

B Hawthorn

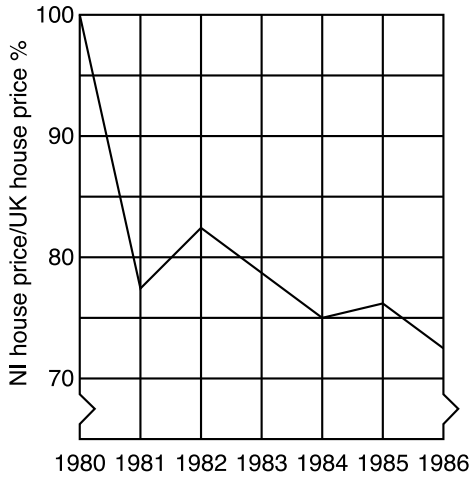
C Holly

D Hornbeam

E Mountain Ash

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19 (Q38) The graph below shows Northern Ireland house prices as a percentage of the UK average (1980=100%) for the period 1980 to 1986.



Which one of the following pieces of information can be inferred from the graph above?

A House prices in Northern Ireland fell by almost 30% in the period shown.

B House prices in Northern Ireland rose in some years and fell in others.

C Average house prices in the UK (outside Northern Ireland) have risen over the period shown.

D In some years average prices in Northern Ireland rose compared to those in the UK as a whole.

E House prices in Northern Ireland had peaked in 1980

20 (Q42) To take part in the National Lottery, entrants must select six different numbers between 1 and 49 inclusive.

My mother asked me to buy a National Lottery ticket for her. "Choose any six numbers you like," she said, "but make sure that each one is a multiple of 3, or contains a 3, or 6 or 9."

How many numbers can I choose from?

A 15
B 16
C 22
D 25
E 31

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21 (Q43) Oliver's first job is with Bubble Engineering Ltd. The company grants paid holidays each year on the following basis:

Length of Employment	*Holiday entitlement*
First 5 years	20 days
Sixth to tenth year	25 days
Eleventh year onwards	30 days
All holidays are granted only at the company's discretion. *Employees who are unable to take their entitlement may carry forward up to 5 days to the following year.*	

Oliver has decided to leave Bubble Engineering on the tenth anniversary of his starting there. Part of his reason for leaving is that he has never been able to take his full holiday entitlement:

Holiday Record

Year	1	2	3	4	5	6	7	8	9	10
Holiday Granted (days)	10	12	15	12	20	18	25	23	20	5 so far

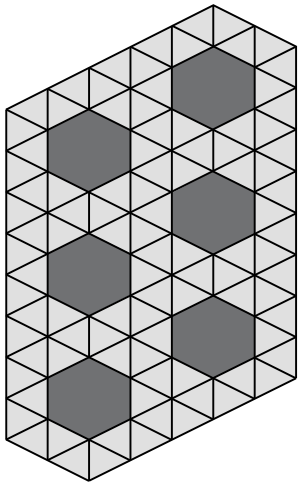
Oliver knows the company always avoids adverse publicity by giving the full entitlement once employees give notice.

How many more days holiday will he get before he leaves?

A 20
B 25
C 30
D 35
E 65

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22 (Q44) The pattern below shows part of the tiling of a large floor in a community centre. It is made up of two types of tile: a hexagon and a triangle.



For a sufficiently large expanse of flooring, approximately what proportion of the two types of tile will be required?

A 1 hexagon to 6 triangles

B 1 hexagon to 9 triangles

C 1 hexagon to 12 triangles

D 1 hexagon to 15 triangles

E 1 hexagon to 18 triangles

23 (Q48) Starting from its depot in town X, a delivery lorry is to make deliveries to three towns A, B and C, in any order which the driver chooses and finally to return to X.

Straight roads connect each town to every other town.

Depending on the layout of the towns, which of the following could **NOT** be the shortest journey for him to take?

A XAXBCBX

B XACBX

C XAXBCX

D XBABXCX

E XABCBAX

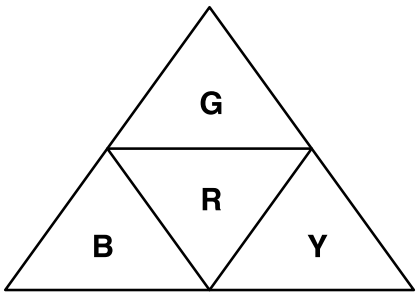
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24 (Q49) A rainwater butt in Joan's garden fills from a flat roof of an equivalent area to 25 square metres. The butt holds 200 litres and is full when Joan starts using it in the summer. During the gardening season, 160 litres falls on each square metre of roof. Only half of this is collected as the rest evaporates. Joan uses 100 litres from the butt to top up her garden pond each week for 25 weeks. If there is not enough water in the butt, she completes the job with mains water which is rather expensive. During the summer months there is never enough rain to overfill the butt.

How much mains water would she expect to use in an average summer?

A 0 litres
B 300 litres
C 500 litres
D 2000 litres
E 2300 litres

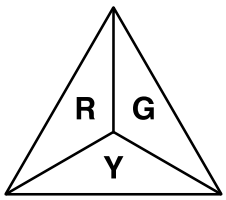
25 (Q50) Four triangles of card coloured on one side are joined as shown.

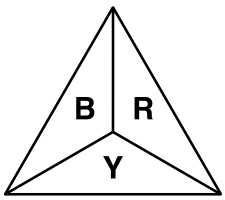


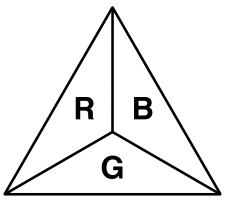
G = Green
R = Red
B = Blue
Y = Yellow

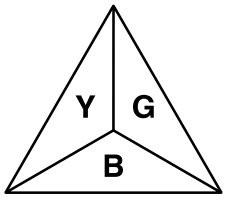
The joined triangles are then folded and joined to make a pyramid.

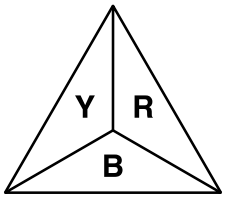
Which of the following could **not** be a view from above?

A


B


C


D


E


END OF TEST

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